Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Owned **fleet orchestration** for multi-cloud compute clusters (AWS, GCP, CoreWeave), writing the scheduling, resource allocation, and **fault-tolerant placement** logic that decides where long-running workloads run.
- Kept fleet utilization up through **node loss** and provider failures by detecting unhealthy capacity and moving work onto healthy nodes, with **Prometheus/Grafana** observability and **autoscaling**.
- Built a **distributed executor** in **Java** and **Python** exposing typed **REST APIs** over Docker and **AWS Lambda**, holding **sub-200ms p99** across millions of daily transactions.
- Designed a content-addressed **S3** artifact store with **Redis** checkpointing so interrupted workflows **resume from the last committed step**, at **99%+ uptime**.

QureAI | *AI Engineer* Mar 2026 – Present

- Cut **p99 latency 66% (420ms to 140ms)** across **50K+ daily requests** by moving synchronous services to **async processing** with horizontal autoscaling.
- Built **CI/CD** with automated quality gates, failure alerting, and **automated rollback** for safe production releases.
- Built an **SRE incident agent** that investigates production alerts and escalates data questions to a SQL agent over **A2A**.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Architected **25+ Java Spring Boot microservices** for financial transaction processing, sustaining **99.9% uptime with zero data loss** for **1M+ users** on **AWS (EC2, RDS, SQS)**.
- Raised service availability to **99.9%** by building health checks, retry policies, and **automated failure recovery** into the serving layer, cutting **p99 latency 32%**.
- Cut production escalations **60%** in one quarter by standardizing retry, timeout, and failure contracts across **25+ services**, driving **root cause analysis** and post-incident reviews.
- Led **architecture and code reviews** on design patterns, reliability, and scaling, and **mentored engineers** on system design and SDLC practices.

Projects

Distributed NoSQL Database `github.com/ganeshhgupta/distributed-kv-store`

- **Durable distributed storage** that confirms writes only after a **majority of nodes** store them, **repairs crashed nodes** by replaying a **write-ahead log**, and resyncs partitioned replicas via **leader-follower replication**.

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system that triages production alerts, routes each to a specialized sub-agent with **tool use**, and writes a structured **root-cause brief**.

Watch-your-LLM `github.com/ganeshhgupta/watch-your-llm`

- Observability platform that traces model calls, latency, and token usage, with a **FastAPI** backend, **PostgreSQL**, and a **React/TypeScript** dashboard.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Aug 2016 – Jul 2020

Skills

Languages: Java, Python, SQL, TypeScript

Performance Engineering: Latency and Throughput Optimization, Bottleneck Diagnosis, Profiling, Async Processing, Caching and Memoization, SQL Query Tuning, Cost and Efficiency

Distributed Systems: Microservices, Fault Tolerance, Autoscaling, Fleet Orchestration, Kafka, SQS, Redis

AWS and Operations: EC2, S3, Lambda, RDS, Docker, Kubernetes, Prometheus/Grafana, CI/CD, On-Call, Root Cause Analysis

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer